RINGS WITH DIVISIBILITY COMPARABLE FOR EVERY PAIR OF ELEMENTS, CHARACTERIZED BY PRINCIPAL FINITELY GENERATED IDEALS WITH A UNIQUE MAXIMAL IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 37637-37649 (canonical tag 0ASN) and the counted proof is the proof environment at source lines 37651-37672, which immediately follows it and is its own proof.

Editorial note. The counted statement keeps the author's own title Generalized valuation rings on the lemma environment and contains the author's reference environment recording the source of the result; the retained preamble defines that environment as comment-type, so its body and the citation inside it are discarded when the document is typeset, which is why the card needs no bibliography.

Editorial note. The proof states that the equivalence of the first and third conditions is straightforward and does not spell it out; the two substantive implications, from the second condition to comparability and from comparability back to the second condition, are both carried out in full inside the counted span, so the card is not an instance of an omitted argument.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which is exactly what makes the reference environment a comment; the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- In a local ring an element whose complement from a relation involving two generators lies in the maximal ideal forces one of the two coefficients to be a unit. used at proof lines 37656-37661. Produces the comparability of divisibility from the principality of the ideal generated by two elements.
- Two distinct maximal ideals can be separated by elements lying in each but not the other, and such elements are mutually nondivisible. used at proof lines 37665-37669. Shows that comparability of divisibility forces the ring to be local.
- If every two elements are comparable by divisibility then every finitely generated ideal is principal by induction, and a valuation ring is an example. used at proof lines 37669-37672. Completes the cycle of equivalences and handles the final statement.

Source reference. Original lemma 00IB: sources/stacks-chow/tags.txt, source lines 673–673. The reference is cited at proof lines 37670-37672 for the final statement,

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that valuation rings satisfy the conditions of the lemma; it is a pointer to a standard characterization and the cycle of equivalences proved here is inside the counted span.

Lemma 0.2 (Generalized valuation rings)**.** *Let R be a nonzero ring. The following are equivalent*

(1) *For $a, b \in R$ either a divides b or b divides a.*
(2) *Every finitely generated ideal is principal and R is local.*
(3) *The set of ideals of R is linearly ordered by inclusion.*

This holds in particular if R is a valuation ring.

Proof. Assume (2) and let $a, b \in R$. Then $(a, b) = (c)$. If $c = 0$, then $a = b = 0$ and a divides b. Assume $c \neq 0$. Write $c = ua + vb$ and $a = wc$ and $b = zc$. Then $c(1 - uw - vz) = 0$. Since R is local, this implies that $1 - uw - vz \in \mathfrak{m}$. Hence either w or z is a unit, so either a divides b or b divides a. Thus (2) implies (1).

Assume (1). If R has two maximal ideals $\mathfrak{m}_i$ we can choose $a \in \mathfrak{m}_1$ with $a \notin \mathfrak{m}_2$ and $b \in \mathfrak{m}_2$ with $b \notin \mathfrak{m}_1$. Then a does not divide b and b does not divide a. Hence R has a unique maximal ideal and is local. It follows easily from condition (1) and induction that every finitely generated ideal is principal. Thus (1) implies (2).

It is straightforward to prove that (1) and (3) are equivalent. The final statement is Algebra, Lemma 00IB. □